THE QUESTION

Does the winner depend only on what people prefer, or also on how they reason and what they can see?

EXPERIMENT 1

Everyone sees everything

Full information · 30 towns · 40 rounds

EXPERIMENT 2

How voters reason

84 voter mixes × 30 towns

EXPERIMENT 3

Who they can hear

5 hearing radii, walking or still, × 30 towns

EXPERIMENT 4

Who tries to sway them

A loyal 10% with 0 to 80 followers × 30 towns

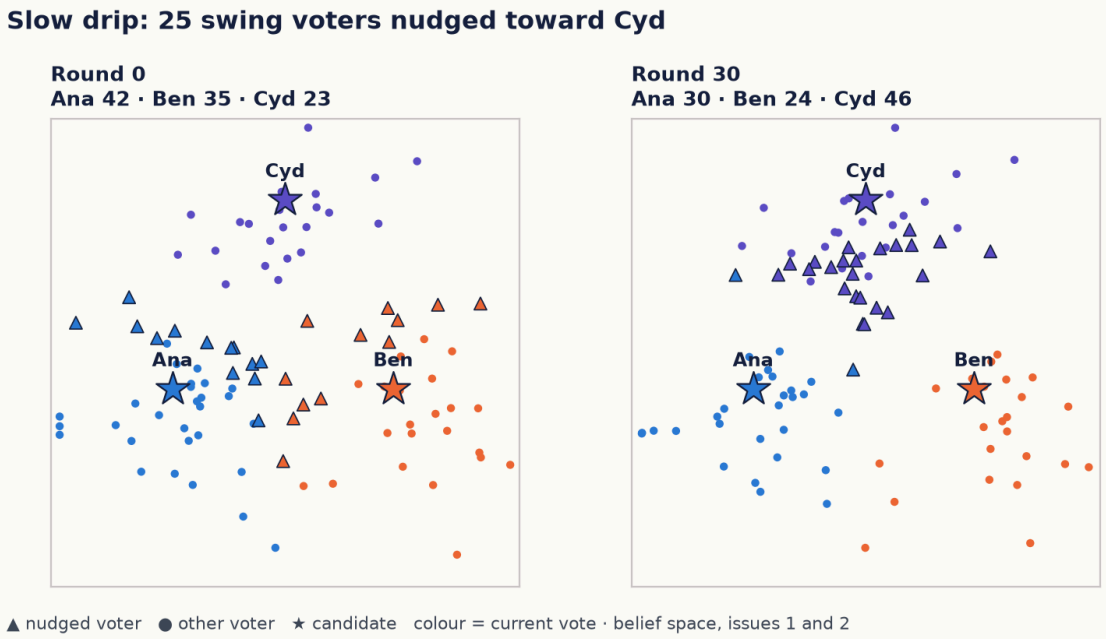
EXPERIMENT 5

Who catches them

A watchdog that sees only vote totals · 10 attack styles

SETUP FOR EXPERIMENT 5

Can a watchdog catch a slow takeover?



One attack, start and end. Triangles are the voters being pushed. Over 30 rounds they drift toward Cyd, and Cyd goes from third (23 votes) to first (46).

THE TOWN

100 people, three candidates: **Ana 42, Ben 35, Cyd 23**. Everyone votes for the candidate closest to their views.

THE ATTACKER

Has a budget to nudge chosen voters' views toward Cyd, a little every round. We tried 10 styles: fast or slow, few or many voters.

THE WATCHDOG

Sees **only the vote totals**, like an election monitor. It wins if it sounds the alarm **before Cyd wins three rounds in a row**.

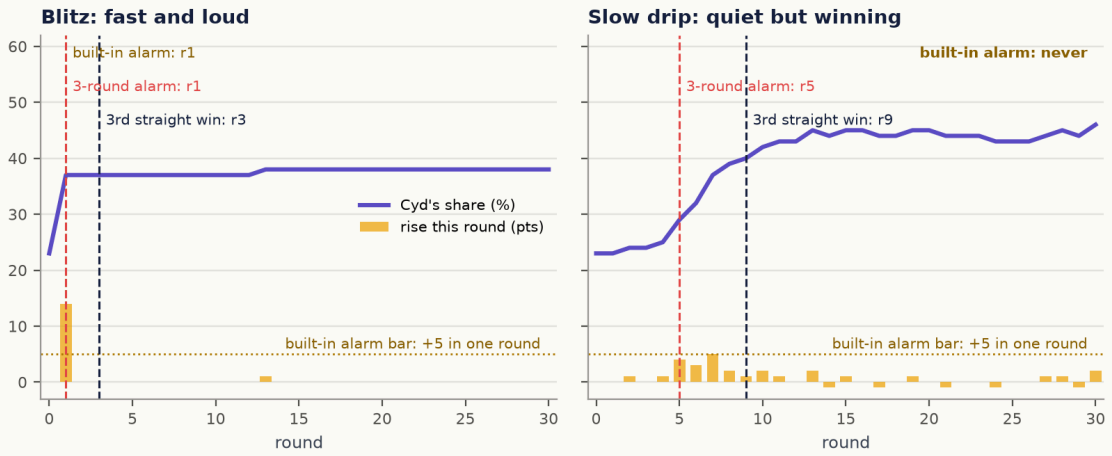
THE BUILT-IN ALARM

Fires if the target jumps more than 5 points in a single round.

Views also drift a little on their own every round · 30 rounds · every attack has an untouched twin town

EXPERIMENT 5

Watch the trend, not the jump



Purple: Cyd's vote share. **Yellow:** how much it rose that round. The blitz jumps at once; the slow drip never rises more than 5 in a round, yet wins by round 9.

WHAT WE GOT

The one-round alarm caught **20 of 25** winning attacks: it misses slow, patient ones. Our best rule, **"alarm if the target rose more than 4 points over 3 rounds"**, caught **25 of 25**, with **no false alarms** in 40 untouched towns.

CAN THE ATTACKER DODGE IT?

We let the attacker try 36 plans. Against the one-round alarm, a slow drip wins every time unseen. Against the 3-round rule, **none** got through.

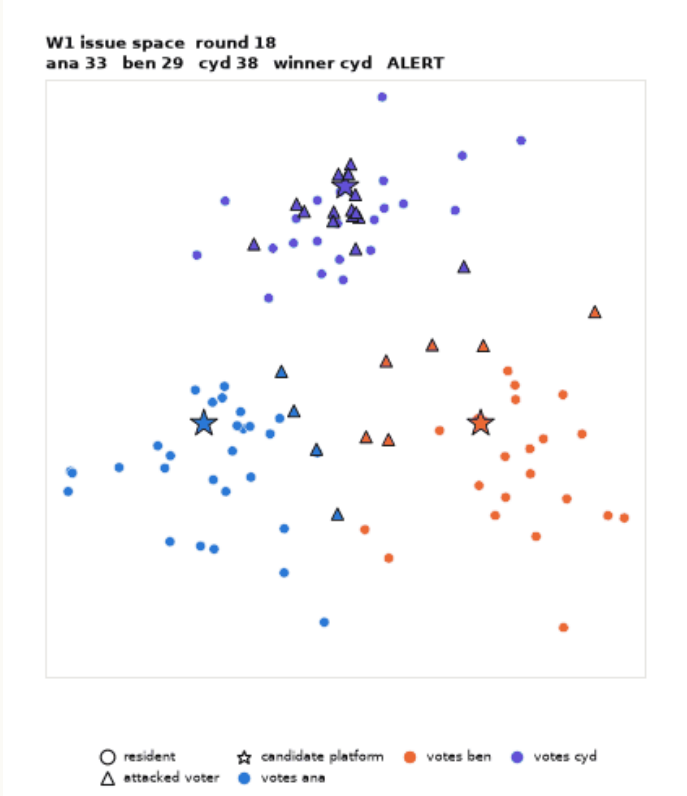
OUR REAL-WORLD READING

Monitors that only look for sudden swings miss patient manipulation. Looking a few rounds back is enough to catch it.

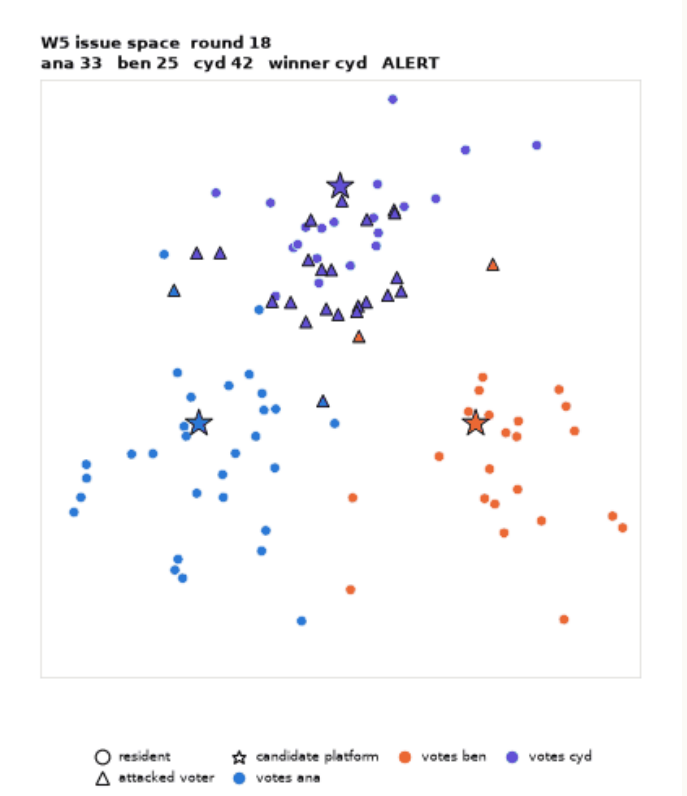
Best rule picked from 101 candidates on one set of towns, then tested on new ones · a 3-point one-round bar also works

EXPERIMENT 5 · WATCH IT HAPPEN

A blitz is loud, a slow drip is quiet



Blitz: Cyd jumps 14 points in round 1. Every alarm fires at once.



Slow drip: Cyd creeps up a few points a round and still takes the lead.

Belief space · ▲ pushed voters · ★ candidates · colour = current vote · 'ALERT' appears when the 3-round alarm fires

TAKEAWAYS

1. With shared information the first leader locks in: in all 30 towns after one round, proved for each.

2. The mix of voters moves the margin, almost never the winner: 19 upsets in 2,520 elections.

3. Changing contacts make the winner unstable, even with fixed beliefs: with 32 heard, Left leads in 29 of 30 towns after round 1, 10 by round 150.

4. A loyal minority with even a small audience can decide the result: with 5 followers each, Centre goes from 2 to 23 of 30 towns.

5. A watchdog must watch the trend, not the jump: a 3-round look-back caught 25 of 25 slow takeovers, with no false alarms.

APPENDIX

Everything behind the results

MATHEMATICAL MODEL

The town as equations

Preferences, hearing, the four decision rules, the parity effect and the full dynamics, in five parts

PROOFS AND THEORY

Why it locks in, why it wanders

The one-step lock-in proof, small-sample odds and stability, and the memory-based voter

EXTRA EXPERIMENT

Campaigns vs echo chambers

Three parties, five campaign playbooks, all 125 combinations

Mathematical slides adapted from Rishi's version · extra experiment run on a separate town of 100 people

APPENDIX · MATHEMATICAL FORMULATION

The model in one slide

Preferences	$n = 300$, $C = \{\text{Left, Right, Centre}\}$: $c \succ_i c' \Leftrightarrow \|\theta_i - \kappa(c)\| < \|\theta_i - \kappa(c')\|$
Hearing set	$H_i(t) = \{\, j \neq i : \|x_i(t) - x_j(t)\| \leq r \,\}$, plus follower links in Experiment 4
Observation	$o_i(t) = \{\, b_j(t-1) : j \in H_i(t) \,\}$: last round's ballots only
Policy	$b_i(t) = \sigma_i(o_i(t))$, $\sigma_i \in \{\text{sincere, copycat, front-runner, pivot-seeker}\}$
Outcome	$s(c, t) = \|\{\, i : b_i(t) = c \,\}\|$, winner $W(t) = \arg\max_c s(c, t)$
Process	$X(t+1) = \Phi(\, X(t), \xi(t) \,)$: a Markov chain whose only randomness is walking

The next five slides spell out each part

Voters, preferences and the plurality rule

Agents, candidates	$N = \{1, \ldots, n\}$, $n = 300$ · $C = \{\text{Left, Right, Centre}\}$, $m = 3$
Preferences	Ideal point θ_i, platform $\kappa(c)$ in $[0,1]^2$: $c \succ_i c' \Leftrightarrow \lVert\theta_i - \kappa(c)\rVert < \lVert\theta_i - \kappa(c')\rVert$
Utility	$u_i(c) = (m - 1 - \text{rank}_i(c)) \,/\, (m - 1) \in \{1, ½, 0\}$; rank 0 = favourite
Ballot, tally	$b_i(t) \in C$, $s(c, t) = \lvert\{\, i : b_i(t) = c \,\}\rvert$, $b_i(0) = \text{top}_i$
Winner	$W(s) = \arg\max_c s(c)$; ties broken by a fixed party order
Town generator	$\theta \sim \text{Normal}((0.25, 0.45), 0.1^2 I)$ for 120 voters, $\text{Normal}((0.64, 0.50), 0.1^2 I)$ for 180; $x_i(0) \sim \text{Uniform}[0,1]^2$; random type · 30 independent towns

Information: who hears whom

Walking	$x_i(t+1) = \text{reflect}(\, x_i(t) + \ell \cdot (\cos\varphi, \sin\varphi)\,)$, $\varphi \sim U[0, 2\pi)$, $\ell \sim U[0, 0.03]$; standing still: no step
Hearing graph	$j \to i$ in $G(t) \Leftrightarrow j \neq i$ and $\lVert x_i(t) - x_j(t) \rVert \leq r$; full information = complete graph
Followers (Exp. 4)	Fixed edges $F \subset \Lambda \times N$: each loyal voter $\lambda \in \Lambda$ gets f uniformly random followers
Hearing set	$H_i(t) = \{\, j : j \to i \text{ in } G(t) \,\} \cup \{\, \lambda : (\lambda, i) \in F \,\}$
Observation	$o_i(t) = \{\, (j, b_j(t-1)) : j \in H_i(t) \,\}$, $n_i(c, t) = \lvert\{\, j \in H_i(t) : b_j(t-1) = c \,\}\rvert$

Hearing size ≈ $(n - 1)\pi r^2$: 9.4 at $r = 0.1$ (measured ≈ 9) · map edges cut big circles: 150 predicted at $r = 0.4$, 104 measured

Decision rules: each agent's policy

Sincere 40% · control	$b_i(t) = \text{top}_i$, the first choice in $\succ_i$, every round
Copycat 20% · local majority	$b_i(t) = \arg\max_c n_i(c, t)$, ties to the earlier candidate in $\succ_i$; sincere if $H_i(t) = \emptyset$
Front-runner 30% · Top-2, next slide	Restrict to the two apparent leaders $T_i(t)$; weigh each by the chance that a tie between them makes your vote decisive
Pivot-seeker 10% · pivotal voter	Belief $\hat{s}_i$ = round-0 tally, updated with every observed switch. Outcome $w_i(c) = W(\hat{s}_i - e(b_i) + e(c))$. Switch to $c^* = \arg\max u_i(w_i(c))$ only if $u_i(w_i(c^*)) > u_i(w_i(b_i))$

Engine policies: sincere, sensed_plurality, top_k (K = 2), pivotal · e(c) = unit vector for candidate c · types assigned independently of opinion

The front-runner rule and a parity effect

1 · Smoothed local estimate	$\hat{p}_i(c) = (n_i(c) + 1) / (\lvert H_i \rvert + m)$
2 · Apparent leaders	$T_i = \{x, y\}$, the two largest $\hat{p}_i$; $q = \hat{p}_i(x) / (\hat{p}_i(x) + \hat{p}_i(y))$
3 · Chance your vote decides	$\pi(x, y) = P[\ \text{Binomial}(n - 1, q) = (n - 1)/2\ ]$
4 · Score and choice	$V_i(x) = \pi(x, y) \cdot (u_i(x) - u_i(y))$; $V_i = -\infty$ outside T_i. Keep b_i if it attains max V_i, else the best by $\succ_i$

Why 300 and 301 voters differ

$n = 300$: 299 others, an odd number, so an exact tie is impossible and $\pi = 0$.

Inertia decides: stay if already on a leader, else take your preferred leader. A **sticky** front-runner.

$n = 301$: $\pi > 0$, so you always take your preferred leader.

Dynamics: one round and the process

ONE ROUND, T → T + 1

1. Move: new positions x(t+1)
2. Rebuild G(t+1) and hearing sets
3. Observe last round's ballots b(t)
4. All decide at once: $b_i(t+1) = \sigma_i(o_i)$
5. Tally s(t+1), winner W

State	$X(t) = (x(t), b(t), \hat{s}(t))$: positions, ballots, pivot beliefs
Transition	$X(t+1) = \Phi(X(t), \xi(t))$, ξ = walking noise: a Markov chain; deterministic if standing still
Game view	$\langle N, A_i = C, o_i, \Phi, u_i(W) \rangle$ with fixed strategies σ_i
Measured over towns k = 1… 30	$\text{wins}(c) = \Sigma_k \, 1[W(s_k(T)) = c]$; switch rate $\rho(t) = \lvert\{ i : b_i(t) \neq b_i(t-1) \}\rvert / n$

APPENDIX · EXPERIMENT 1 · WHY IT STOPS

One-step lock-in under full information

Proposition

Full information, $n = 300$. If $s(1)$ has the same leader and the same top two as $s(0)$, with gaps of at least 2 so that no single ballot changes either, then $b(2) = b(1)$, and so $b(t) = b(1)$ for all $t \geq 1$.

Checked in all 30 towns: round 1 keeps round 0's top two (Left–Right in 27, Left–Centre in 3).

PROOF, TYPE BY TYPE

Sincere	Ballot never changes.
Copycat	Copies the leader; same leader, same vote.
Front-runner	With $\pi = 0$ it depends only on T; its round-1 vote is already in T, so it stays.
Pivot-seeker	Gap ≥ 2: no vote is decisive, so it stays.

Same ballots give the same observations, so by induction the profile is a fixed point.

Not automatic: with near-ties the same rules can cycle forever, which is why the condition is checked town by town

APPENDIX · EXPERIMENT 3 · WHY

Why: small samples, moving samples

Local samples

P(Left is the strict leader among k heard), round-0 shares:

k	2	9	32	104
P	16%	41%	52%	63%

Most copycats see another leader, so Left's bandwagon never forms.

Standing still

$X(t+1) = \Phi(X(t))$ on a finite set: 3^n ballot profiles × bounded beliefs.

So every run is eventually periodic, guaranteed. We observe period 1 or 2, as for majority rules on symmetric graphs (Goles & Olivos 1980).

Walking

Random contacts: no absorbing profile need exist.

On a finite grid the chain is finite, so the share of rounds each candidate leads converges almost surely (ergodic theorem). The stable object is a frequency, not a winner.

Probabilities: multinomial with median round-0 shares 118.5 / 98 / 83 of 300 · finite-grid result: the engine's stochastic-stability track

APPENDIX · ONGOING ENGINE WORK

A voter with memory and a fading poll

Recent memory, K = 10	$q_i(c)$ = share of c among the latest ballots of everyone met in the last K rounds
Poll every L = 10 rounds	fading weight $w(t) = (L - d + 1)/L$, $d = 1 + ((t-1) \bmod L)$: 1, 0.9, ..., 0.1
Blended estimate	$\hat{p}_i(c) = w \cdot P(c)/\Sigma P + (1-w) \cdot q_i(c)$
Decision	$F(a) = W((n-1)\hat{p}_i + e(a))$; vote for your favourite reachable winner c* only if $c^* \succ_i F(b_i)$

Convergence

A 4-voter cycle refutes the original version; no guarantee for n ≥ 100.

Finite grid

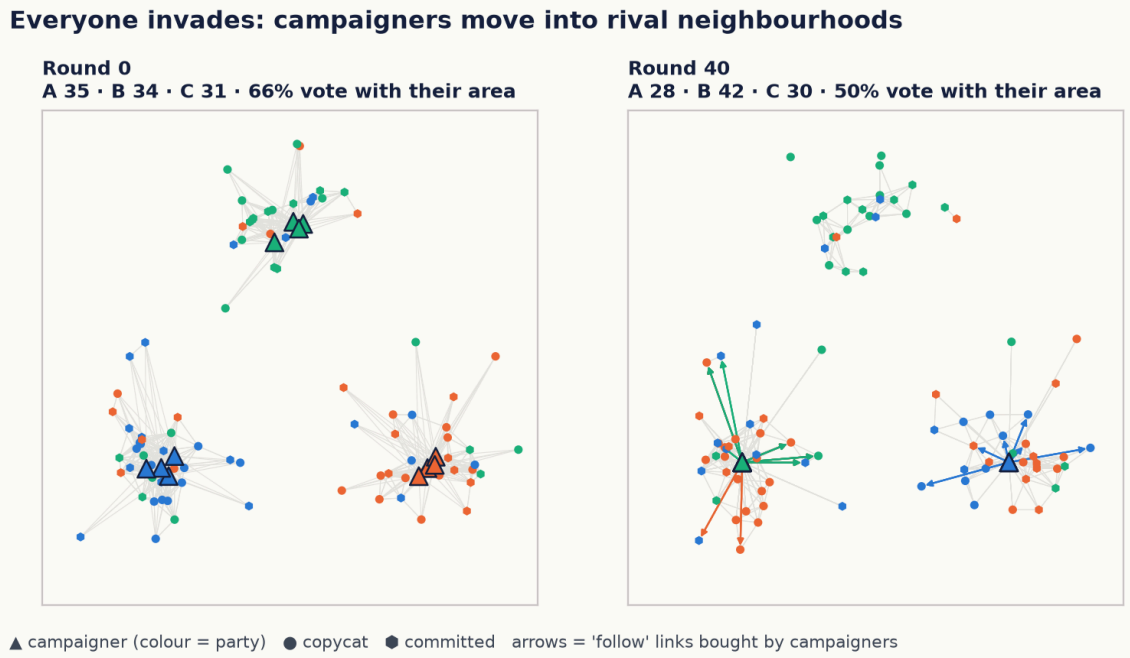
On a 41 × 41 lattice, long-run frequencies converge almost surely.

Pilot, 200 voters, 5 seeds

Share error, moving town: poll 0.024, blend 0.085, memory alone 0.167.

APPENDIX · EXTRA EXPERIMENT · SETUP

Campaigners who change who you hear



Everyone invades. Campaigners walk into rival neighbourhoods and get voters to follow them. By round 40 only 50% still vote with their neighbourhood.

THE TOWN

Three neighbourhoods, one per party: **A 34, B 30, C 24**. Most residents are copycats: they vote like the 4 or 5 neighbours they see, so each area becomes an **echo chamber**.

THE CAMPAIGNERS

4 per party. They move fast, can poll a wide area, and can get up to 6 voters to **follow** them, so those voters see the campaigner's vote.

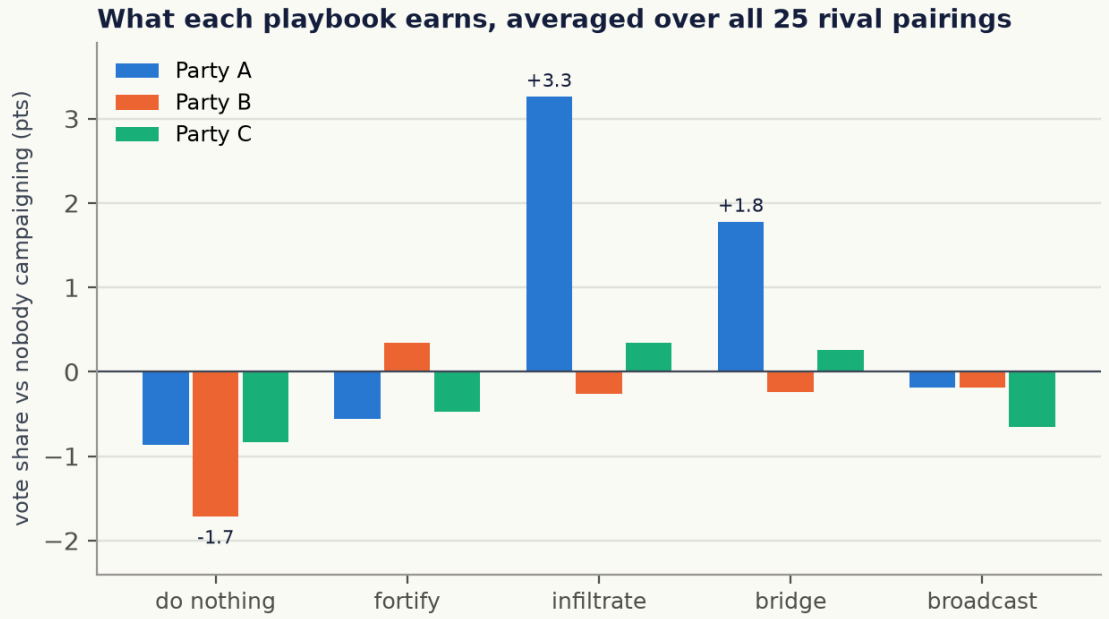
THE PLAYBOOKS

Do nothing · **defend** your home area · **invade** the biggest rival area · **work the border** · **random** followers. Every party picks one.

All 125 combinations of playbooks × 3 towns × 40 rounds · score: a party's vote share over the last 10 rounds

APPENDIX · EXTRA EXPERIMENT

Attack if you lead, defend if you chase



Each bar: how much a playbook earns a party on average, against every combination of its rivals' playbooks. “Fortify” is defend; “infiltrate” is invade.

WHAT WE GOT

A (biggest) should **invade**: +3.3 points. **B** should **defend**; doing nothing costs it 1.7. **C** (smallest) can't move the needle and won **0 of 375** elections.

DOES ANYONE SETTLE?

No. Each party keeps switching to beat the others, in a loop. The closest to stable is **everyone invading**.

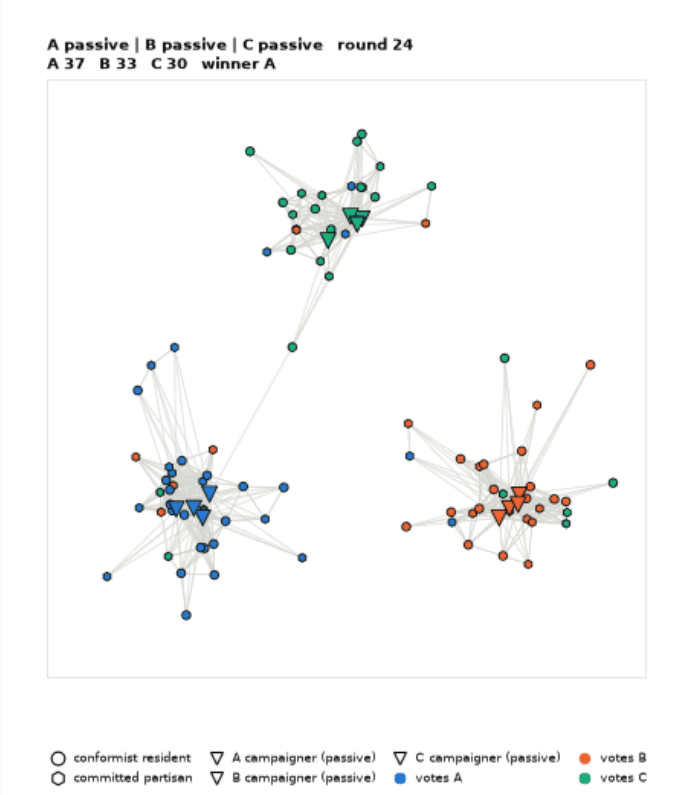
OUR REAL-WORLD READING

Campaign wars break echo chambers (79% → 55% vote with their area) but barely change who wins.

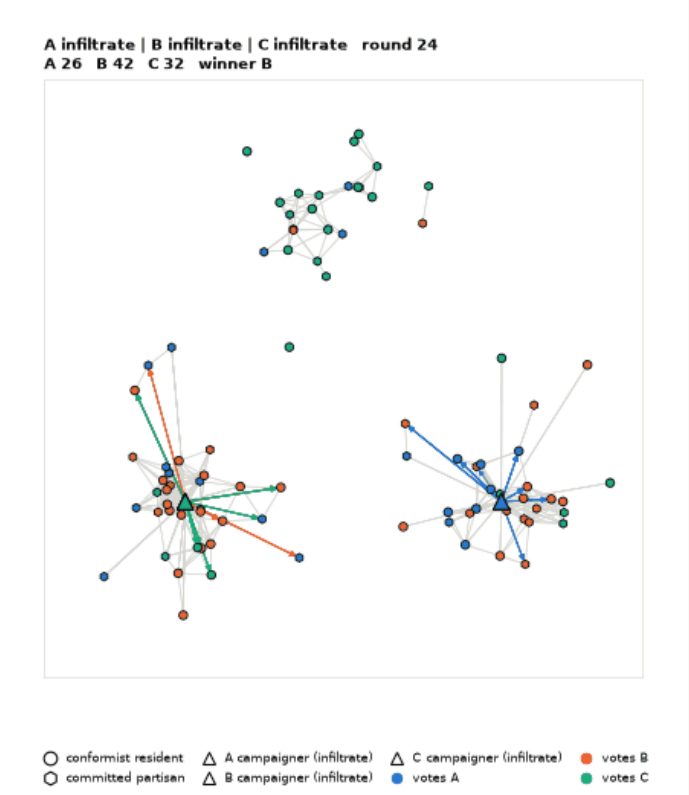
Across all 375 elections: A wins 348, B 27, C 0 · B's only upset route is working the border while A stays home

APPENDIX · EXTRA EXPERIMENT · WATCH IT HAPPEN

Nobody campaigns vs everyone invades



Nobody campaigns: neighbourhoods seal themselves and A wins.



Everyone invades: follow links tear A's area open and B takes the lead.

Town map · ▲ campaigners · ● copycats · ⬢ committed · arrows = follow links bought by campaigners